

CSE 150A-250A AI: Probabilistic Models

Lecture 14

Fall 2025

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Slides adapted from previous versions of the course (Prof. Lawrence, Prof. Alvarado, Prof Berg-Kirkpatrick)

Agenda

Review

Reinforcement Learning

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EM algorithm for HMMs

- CPTs to re-estimate:

$$\pi_i = P(S_1=i)$$

$$a_{ij} = P(S_{t+1}=j|S_t=i)$$

$$b_{ik} = P(O_t=k|S_t=i)$$

- E-step in HMMs must compute:

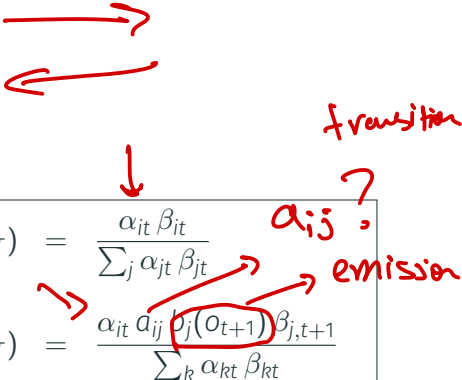
$$P(S_1=i|o_1, o_2, \dots, o_T)$$

$$P(S_{t+1}=j, S_t=i|o_1, o_2, \dots, o_T)$$

$$\left[\underbrace{P(O_t=k, S_t=i|o_1, o_2, \dots, o_T)}_{\text{special case of below (t=1)}} \right] = \underbrace{l(o_t, k)}_{\text{special case of below (t=1)}} \underbrace{P(S_t=i|o_1, o_2, \dots, o_T)}$$

Forward-backward algorithm for inference in HMMs

- Summary of E-step:


$$P(S_t=i|o_1,\dots,o_T) = \frac{\alpha_{it} \beta_{it}}{\sum_j \alpha_{jt} \beta_{jt}}$$
$$P(S_t=i, S_{t+1}=j|o_1,\dots,o_T) = \frac{\alpha_{it} a_{ij} \underbrace{b_j(o_{t+1})}_{\text{emission}} \beta_{j,t+1}}{\sum_k \alpha_{kt} \beta_{kt}}$$

Handwritten annotations:
- a_{ij} : transition
- $b_j(o_{t+1})$: emission

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- M-step updates:

$$\pi_i \leftarrow P(S_1=i|o_1, o_2, \dots, o_T)$$

$$a_{ij} \leftarrow \frac{\sum_t P(S_{t+1}=j, S_t=i|o_1, o_2, \dots, o_T)}{\sum_t P(S_t=i|o_1, o_2, \dots, o_T)}$$

$$b_{ik} \leftarrow \frac{\sum_t I(o_t, k) P(S_t=i|o_1, o_2, \dots, o_T)}{\sum_t P(S_t=i|o_1, o_2, \dots, o_T)}$$

(for one sequence of observations)

Time complexity of HMM computations

T	length of observation sequence $(o_1, o_2, \dots, o_T)$
n	cardinality of state space $s_t \in \{1, 2, \dots, n\}$
m	cardinality of observation space $o_t \in \{1, 2, \dots, m\}$

- All of the following computations are $O(n^2T)$:

- (a) computing the likelihood $P(o_1, o_2, \dots, o_T)$ *forward*
- (b) decoding $\operatorname{argmax}_{s_1, \dots, s_T} P(s_1, \dots, s_T | o_1, \dots, o_T)$ *viterbi*
- (c) re-estimating $\{\pi_i, a_{ij}, b_{ik}\}$ by **one update of EM** *forward-backward*
- (d) updating beliefs $P(S_t = i | o_1, \dots, o_t)$ **for T steps** *!*

Reinforcement Learning

Reinforcement learning (RL)

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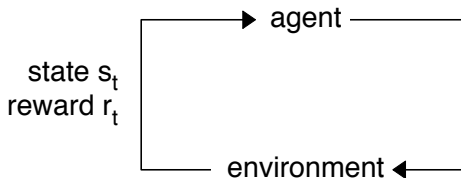
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Agents are actors of any kind: they can be *embodied* in the physical world or *embedded* in a virtual environment.

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5. How to navigate complex worlds with tractable models?
6. How to prove computational guarantees
(e.g., convergence, optimality, efficiency)?

A probabilistic framework for RL

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A probabilistic framework for RL



How do we formalize this process?

A probabilistic framework for RL



How do we formalize this process?

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We define a **Markov decision process**.

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$$\text{MDP} = \{\mathcal{S}, \mathcal{A}, P(s'|s, a), R(s, s', a)\}$$

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2. Transition probabilities are constant over time: *Stationary.*

$$P(S_{t+1}=s' | S_t=s, A_t=a) = \underbrace{P(S_{t+1+\tau}=s' | S_{t+\tau}=s, A_{t+\tau}=a)}_{\text{shifted by } \tau}$$

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Simplifications for CSE 150A/250A

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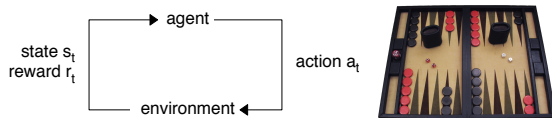
$$\text{MDP} = \{\mathcal{S}, \mathcal{A}, P(s'|s, a), \underline{R(s)}\}$$

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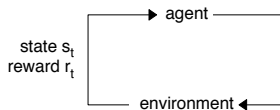
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$$s \in \mathcal{S}$$



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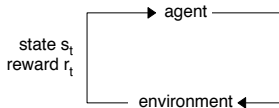
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board position and results of roll of dice

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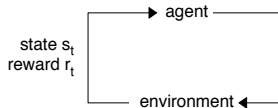


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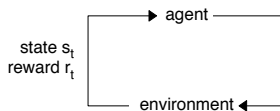
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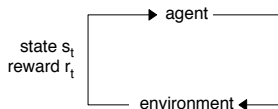


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$R(s) =$

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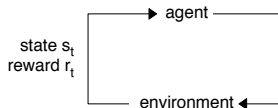


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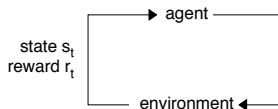


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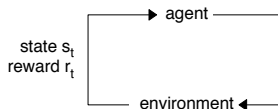


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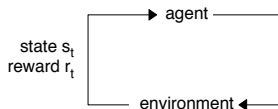
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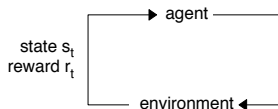
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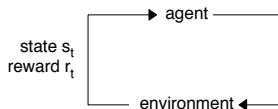
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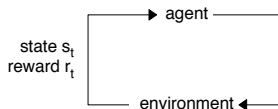
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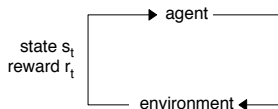
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The Game (Nim):

- Start with 6 objects
- On your turn, you can take 1 or 2 objects.
- Last person to take an object **loses**.

During the game:

- Agent plays first. When it is the agent's turn, draw a slip from the correct state.
- Leave the drawn slip next to that state.

After the game ends:

- **If the agent won:** fold and put all drawn slips back into their states.
- **If the agent lost:** discard the slip for the agent's last action. If a state would become empty, keep its last slip and instead discard the slip from the previous agent action. Return all earlier slips to their states.

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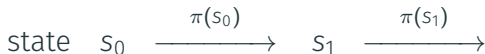
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In this class we will only consider deterministic policies.

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If there are $|\mathcal{A}|$ possible actions in each of $|\mathcal{S}|$ states,
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Decision-making in MDPs

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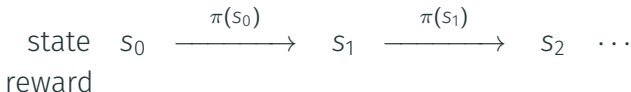
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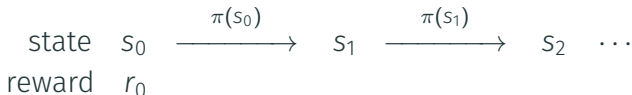
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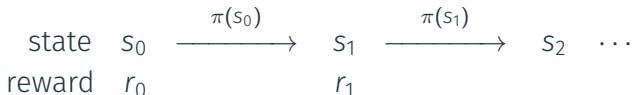
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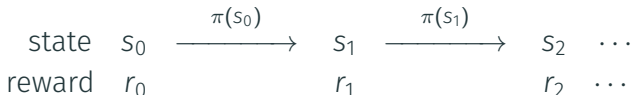
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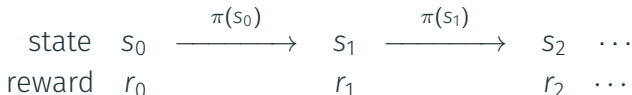
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Transitions occur with probabilities $P(s'|s, \pi(s))$.

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But they are **not** the most commonly used in practice ...

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When γ is close to 1, future rewards are lightly discounted.

These returns can only be optimized by **far-sighted agents**.

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Discounted returns lead to simple iterative algorithms with strong guarantees of convergence.

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Maximizing the expected return is:

- generally wiser than maximizing the best-case return,
- but not as robust as minimizing the worst-case return.

Test your understanding

- Learning from experience in the world



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Which of the following is **NOT** part of the definition of an MDP?

- A. A set of states $\mathcal{S}$ with $s \in \mathcal{S}$
- ☒ B. Transition probabilities: $P(s'|s) = P(S_{t+1}=s'|S_t=s)$
- C. An action space $\mathcal{A}$ with actions $a \in \mathcal{A}$
- D. Reward function $R(S)$
- E. None of the above.

Test your understanding

When calculating the long-term value of a state sequence over time, rewards for states in the near future count more than rewards for states in the distant future.

True (A) or False (B)?

That's all folks!